

QUINCY, IL, USA

WMO: 724430

Lat: 39.937N

Lon: 91.192W

Elev: 769

StdP: 14.29

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 93989

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -0.3	(c) 4.9	(d) -8.6	(e) 3.6	(f) 0.5	(g) -3.9	(h) 4.6	(i) 6.4	(j) 28.2	(k) 26.2	(l) 25.9	(m) 25.1	(n) 9.3	(o) 310	(p) 0.526

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 18.7	(c) 93.1	(d) 76.4	(e) 90.4	(f) 75.4	(g) 88.0	(h) 74.2	(i) 79.0	(j) 89.1	(k) 77.5	(l) 87.2	(m) 75.9	(n) 84.9	(o) 10.1	(p) 220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
76.0	139.9	85.4	74.5	132.7	83.7	73.0	125.8	81.8	43.1	89.6	41.4	87.1	39.9	85.0	85.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
24.3	20.8	18.8	-7.2	97.6	5.7	3.3	-11.3	100.0	-14.6	101.9	-17.7	103.8	-21.9	106.2		
			-7.8	81.4	5.5	1.8	-11.8	82.7	-15.0	83.7	-18.0	84.7	-22.0	86.0		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	53.3	26.6	31.1	42.0	53.5	63.7	72.8	76.3	74.8	67.4	55.5	42.8	31.6	
	DBStd	19.73	12.35	12.20	11.43	9.62	8.38	6.29	5.71	5.36	7.89	9.59	10.31	11.10	
	HDD50	2531	728	537	301	70	5	0	0	0	1	52	262	575	
	HDD65	5448	1191	948	715	362	130	12	1	2	64	322	667	1034	
	CDD50	3734	2	9	54	175	429	684	816	768	524	222	46	5	
	CDD65	1174	0	0	2	16	89	246	352	305	137	26	1	0	
	CDH74	10643	0	0	15	150	693	2176	3505	2677	1218	205	4	0	
	CDH80	3873	0	0	0	21	168	779	1443	997	421	44	0	0	
Wind	WSAvg	9.2	10.6	10.5	10.9	10.9	9.2	8.1	7.0	6.8	7.5	9.0	10.1	9.9	
Precipitation	PrecAvg	39.1	1.5	1.7	3.0	3.9	4.6	3.9	4.2	3.6	4.2	3.3	3.0	2.3	
	PrecMax	66.6	6.1	5.2	11.5	7.8	13.3	8.5	13.2	10.8	10.8	10.8	11.2	8.6	
	PrecMin	20.4	0.1	0.2	0.9	1.0	0.4	0.4	0.1	0.3	0.4	0.4	0.3	0.2	
	PrecStd	9.8	1.3	1.2	1.9	1.8	2.4	2.1	2.6	2.3	2.7	2.3	2.2	1.6	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	61.2	66.6	76.1	83.1	88.3	93.5	99.3	96.4	92.5	85.0	73.1	63.8	
		MCWB	55.0	55.2	62.1	65.8	72.4	75.0	76.5	76.0	73.5	69.1	60.4	58.4	
	2%	DB	54.3	60.2	70.9	78.7	84.2	90.4	93.6	91.8	88.7	79.5	67.8	57.4	
		MCWB	49.1	51.9	58.7	62.7	69.9	74.9	77.8	76.1	71.9	65.1	58.0	51.3	
	5%	DB	48.4	54.1	64.7	74.3	81.4	87.9	90.7	88.9	85.1	75.0	63.2	51.8	
		MCWB	43.3	48.0	55.3	60.9	67.9	74.1	76.9	74.8	71.2	63.6	55.4	47.1	
	10%	DB	42.9	48.2	59.9	70.0	78.0	84.5	87.8	85.6	81.2	70.9	58.6	46.9	
		MCWB	38.4	42.5	51.8	58.2	65.8	71.9	75.1	73.5	68.1	61.4	51.9	42.5	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	56.1	58.7	64.1	68.8	74.9	78.8	81.5	80.8	76.3	71.6	63.5	59.4
			MCDB	59.8	63.7	73.1	79.0	84.8	88.7	91.2	90.4	87.0	81.9	69.3	63.3
2%		WB	50.0	52.9	60.4	65.2	72.2	76.9	79.6	78.4	74.4	68.7	59.7	53.0	
		MCDB	53.4	57.9	68.1	75.4	81.5	86.9	90.2	87.9	84.7	75.7	65.5	56.8	
5%		WB	43.9	47.9	57.0	62.7	70.0	75.2	78.1	76.9	72.7	65.8	56.4	47.4	
		MCDB	48.5	54.5	63.9	71.5	78.5	84.8	88.1	85.9	81.8	72.3	61.9	50.6	
10%		WB	38.8	42.9	52.2	59.8	67.9	73.7	76.5	75.1	70.9	62.7	52.7	42.8	
		MCDB	42.7	48.3	58.5	68.7	75.9	82.7	85.5	82.9	78.6	69.8	58.1	46.7	
Mean Daily Temperature Range	5% DB	MDBR	15.2	16.2	18.3	19.9	18.4	18.1	18.7	18.8	20.9	19.3	17.1	14.4	
		MCDBR	21.9	23.8	24.1	24.6	21.1	20.5	21.1	22.0	23.1	23.3	22.1	20.0	
		MCWBR	17.4	17.5	14.8	14.2	10.9	9.5	8.7	9.3	9.8	12.2	15.4	16.0	
	5% WB	MCDBR	20.3	23.1	22.0	21.2	19.0	18.6	19.0	18.9	19.8	19.0	20.2	18.7	
		MCWBR	17.0	17.9	14.7	13.9	10.8	9.6	9.1	9.3	9.5	12.0	15.7	16.6	
Clear-Sky Solar Irradiance	taub	0.295	0.317	0.340	0.390	0.431	0.429	0.450	0.443	0.395	0.352	0.303	0.289		
	taud	2.455	2.382	2.405	2.263	2.210	2.268	2.194	2.230	2.316	2.441	2.518	2.489		
	Ebn at Noon	278	286	290	280	270	270	263	261	267	270	273	272		
	Edh at Noon	25	31	34	42	45	43	45	43	37	29	23	23		
All-Sky Solar Radiation	RadAvg	610	887	1182	1538	1768	1998	1990	1748	1511	1030	696	521		
	RadStd	70	81	86	114	133	177	122	95	99	115	61	63		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+220	+146
Regional (32 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+184	+126

Nomenclature: See separate page